

Assignment 12

$$r(\theta) = \theta^{\frac{1}{8}}$$

$$\theta \sim U[0,1]$$

a) Who works?

$$\{\theta: r(\theta) = \theta^{\frac{1}{8}} \leq \theta\} = \{0, 1\}$$

b) Since the firm does not observe θ it is impossible for them to enforce choosing the "right" wage in this setting, any lower type being offered a lower wage will always choose the high wage since they are maximizers.

c) Who works for a wage w ?

$$\Theta(w) = \{\theta: r(\theta) \leq w\} = \begin{cases} [0, w^8] & \text{for } w \in [0,1] \\ [0,1] & \text{for } w > 1 \end{cases}$$

$$\mathbb{E}[\theta | \theta \in \Theta(w)] = \begin{cases} \frac{w^8}{2} & \text{for } w \in [0,1] \\ \frac{1}{2} & \text{for } w > 1 \end{cases}$$

In eq we must have

$$\mathbb{E}[\theta | \theta \in \Theta(w)] = w$$

$$\frac{w^8}{2} = w$$



Ligningen løses for w vha. WordMat.

$$w = 0 \quad \vee \quad w = 2^{\frac{1}{7}}$$

2 possible eq

$$w' = 2^{\frac{1}{7}}$$

$$\Theta(w') = [0,1]$$

$$\mathbb{E}[\theta | \theta \in [0,1]] = \frac{1}{2} \neq 2^{\frac{1}{7}}$$

Thus this is not an eq

$$w' = 0$$

$$\Theta(w') = \{0\}$$

$$\mathbb{E}[\theta | \theta \in \{0\}] = 0 = w'$$

Thus this is an eq

d) slides say that the highest wage eq is constrained pareto optimal so

$$w' = 0, \Theta(w') = \{0\}$$

Is constrained pareto optimal

e)

$$r(\theta) = \theta^8$$

$$\theta \sim U[0,1]$$

$$\{\theta: r(\theta) = \theta^8 \leq \theta\} = [0,1]$$

Everyone works

f)

$$\Theta(w) = \{\theta: r(\theta) \leq w\} = \begin{cases} \left[0, w^{\frac{1}{8}}\right] & \text{for } w \in [0,1) \\ [0,1] & \text{for } w \geq 1 \end{cases}$$

$$E[\theta | \theta \in \Theta(r)] = \begin{cases} \frac{w^{\frac{1}{8}}}{2} & \text{for } w \in [0,1) \\ \frac{1}{2} & \text{for } w \geq 1 \end{cases}$$

In eq we must have

$$E[\theta | \theta \in \Theta(r)] = w$$

$$\frac{w^{\frac{1}{8}}}{2} = w$$



Ligningen løses for w vha. WordMat.

$$w = 0 \quad \vee \quad w = \frac{1}{2^7}$$

We check both EQ

" $w = 0$ "

$$\Theta(0) = \{0\}$$

$$E[\theta | \theta \in \{0\}] = 0$$

Thus a EQ

" $w = \frac{1}{2^{8/7}}$ "

$$\Theta\left(\frac{1}{2^7}\right) = \left[0, \frac{1}{2^{\frac{1}{7}}}\right]$$

$$E\left[\theta \middle| \theta \in \left[0, \frac{1}{2^{\frac{1}{7}}}\right]\right] = \frac{1}{2} \cdot \frac{1}{2^{\frac{1}{7}}} = \frac{1}{2^{8/7}} = w$$

g) slides say that the highest wage eq is constrained pareto optimal so

$$w = \frac{1}{2^7}, \Theta(w) = \left[0, \frac{1}{2^{\frac{1}{7}}}\right]$$

Is constrained pareto optimal